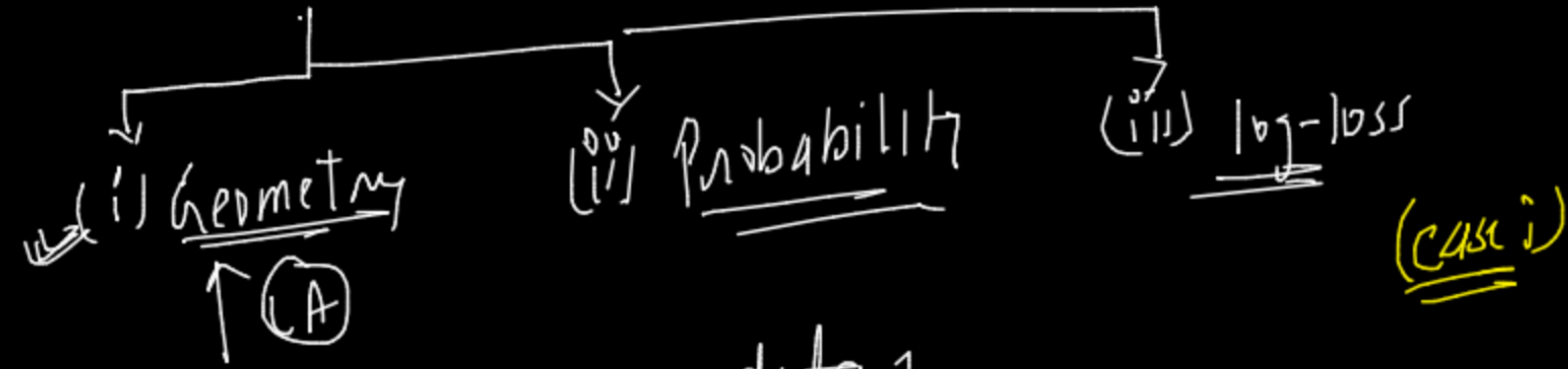


⇒ Logistics Regression

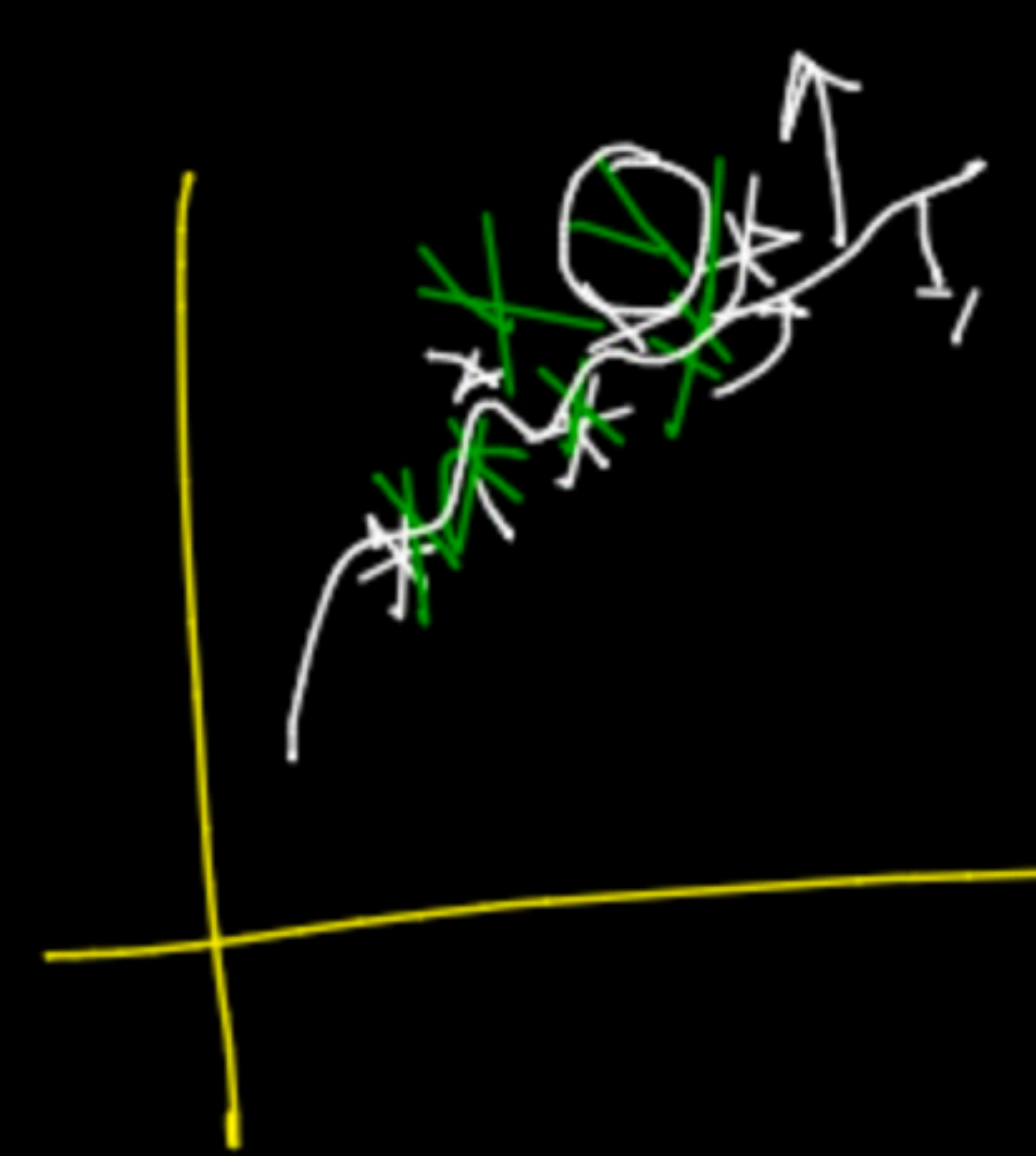
↓
'Classification Problems' →



Assumption →

data 1
→ (i) linearly separable

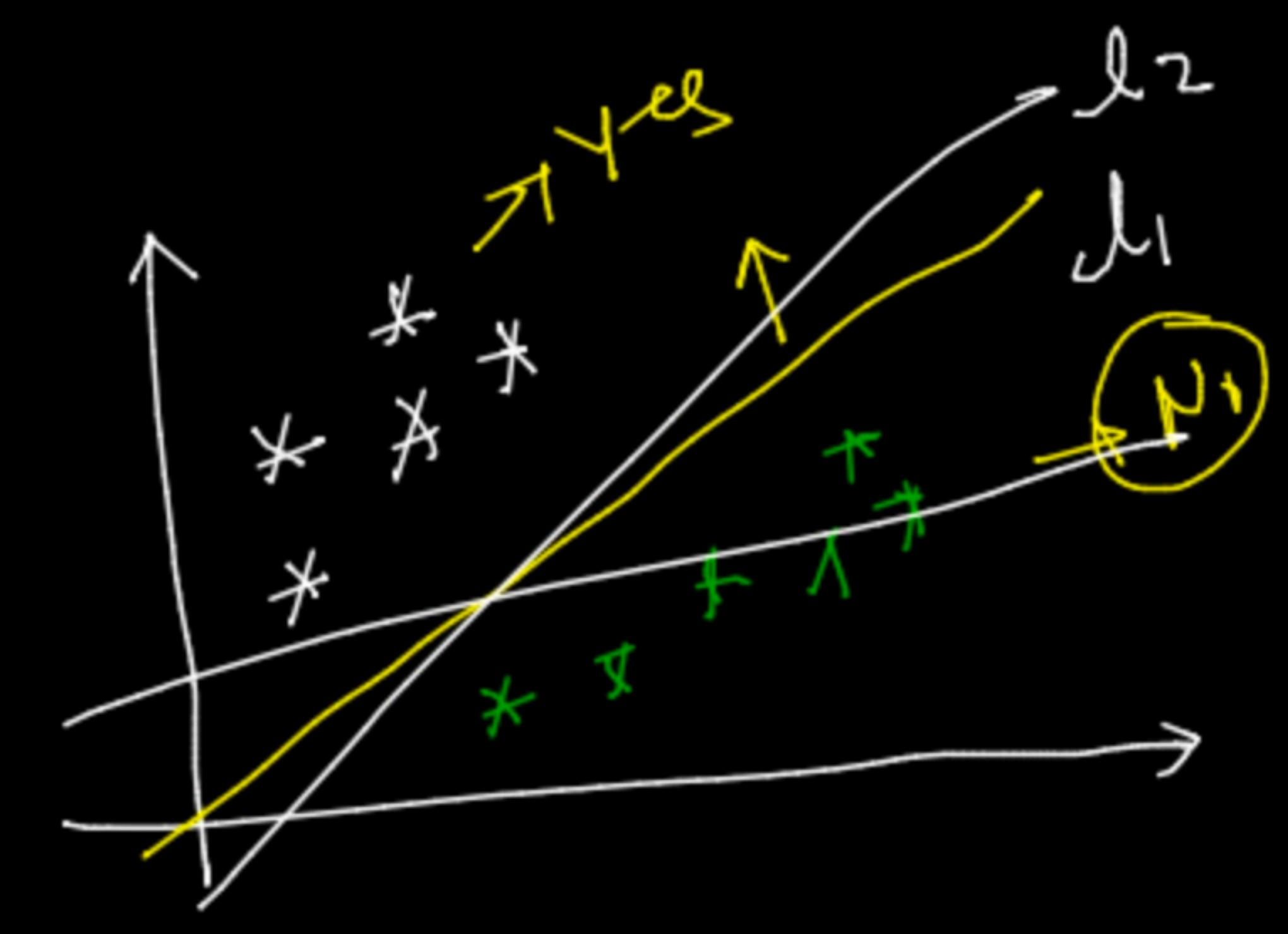
(case ii)



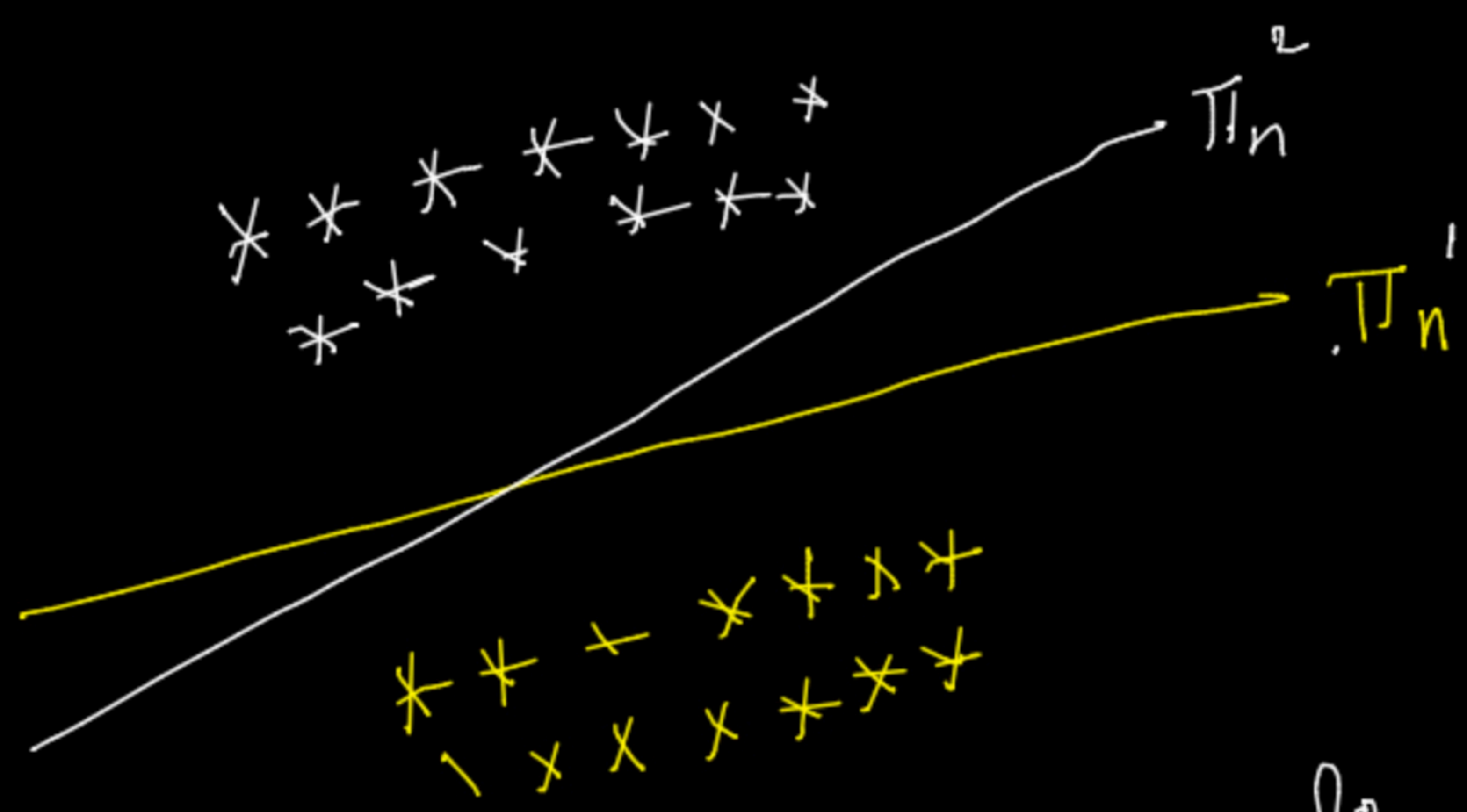
(2D)
↓
(line)



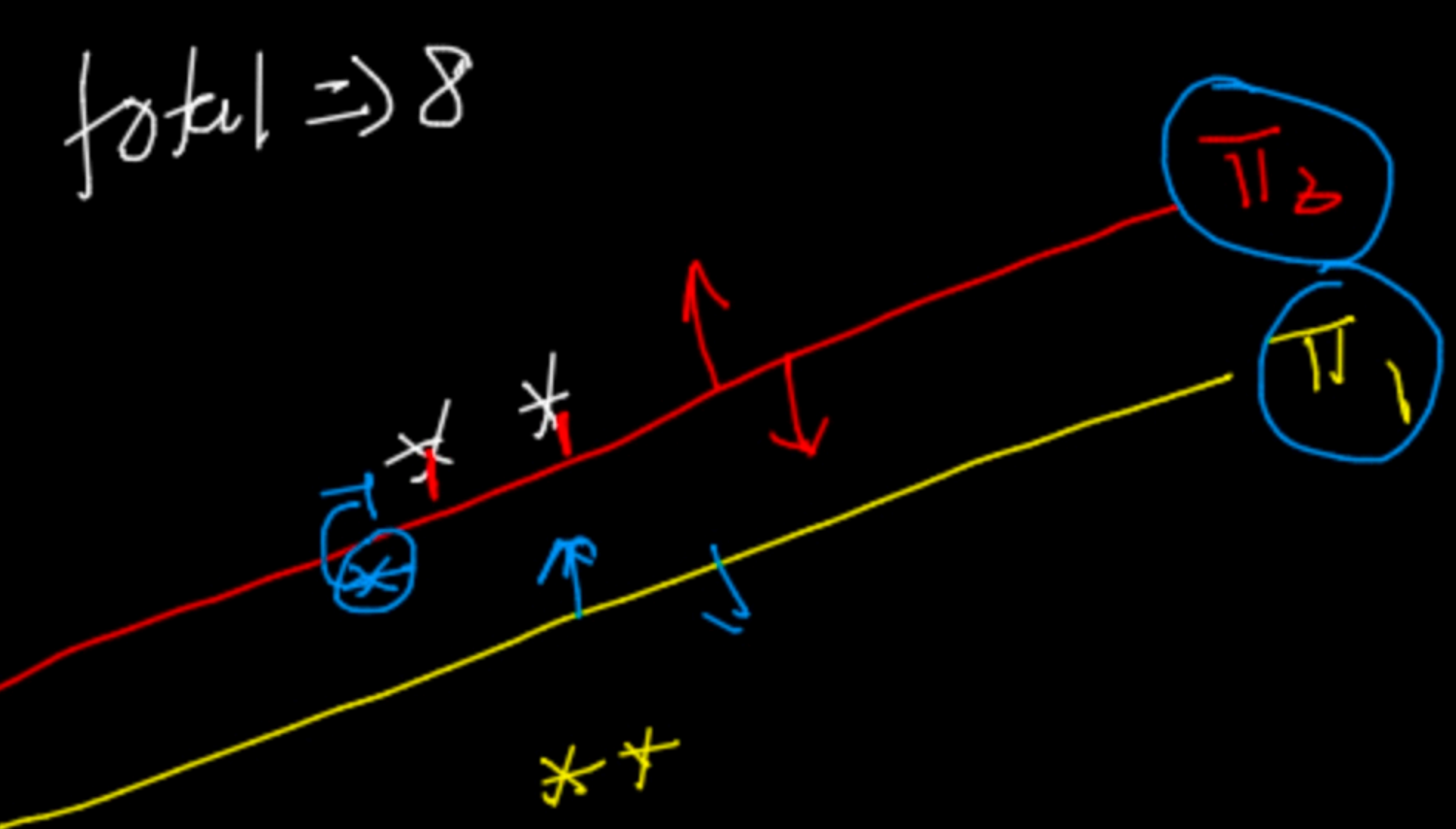
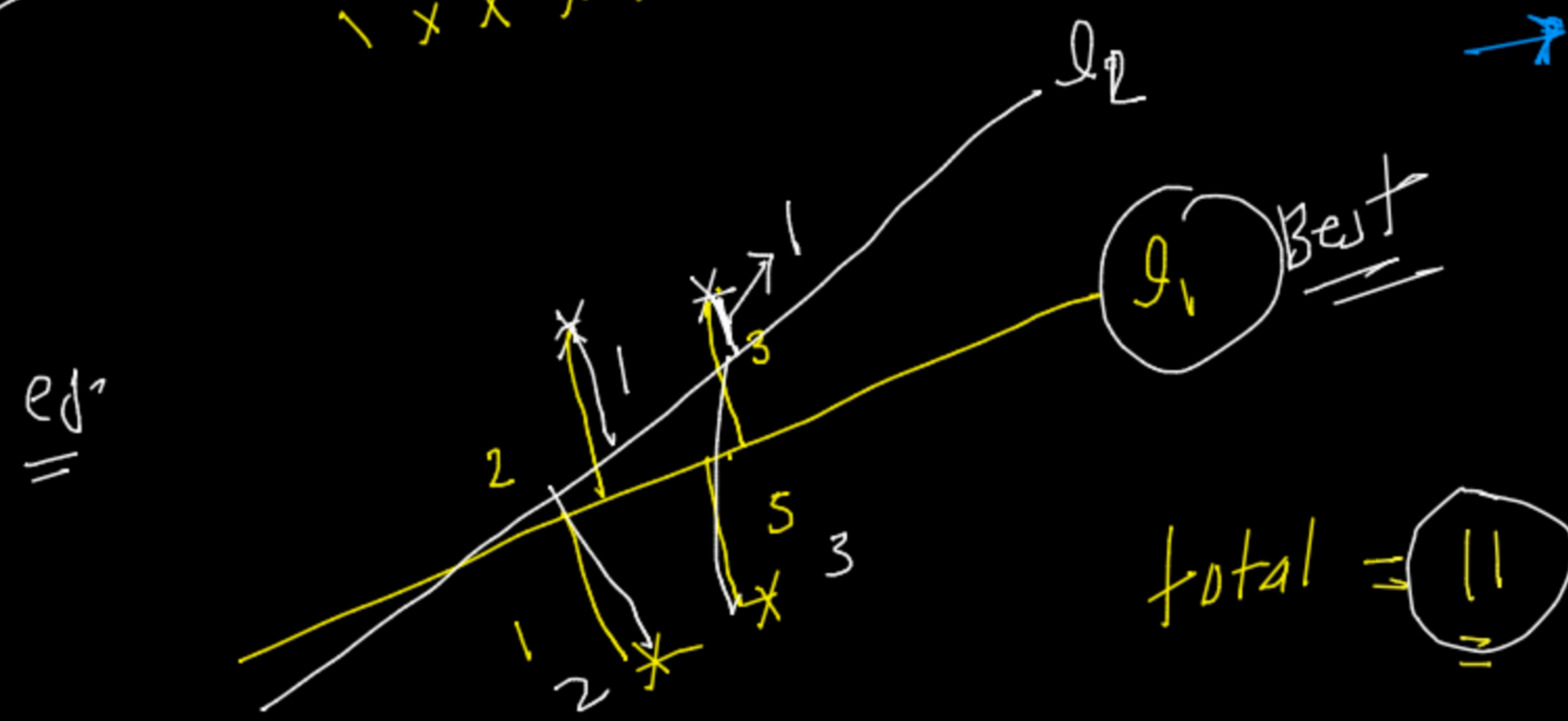
(EAA) - (ims)



f_1	T
b	b



L_2
 $\downarrow$
Objective $\rightarrow$ Best fit line / plane $\rightarrow$
 which can maximize the distance
 from plane / line



Equation of plane

$$\underline{y = mx + c}$$

$$\underline{\pi} = \underline{w^T} \cdot x + b$$

$$[0, 0.1, 0.2] [- -]$$

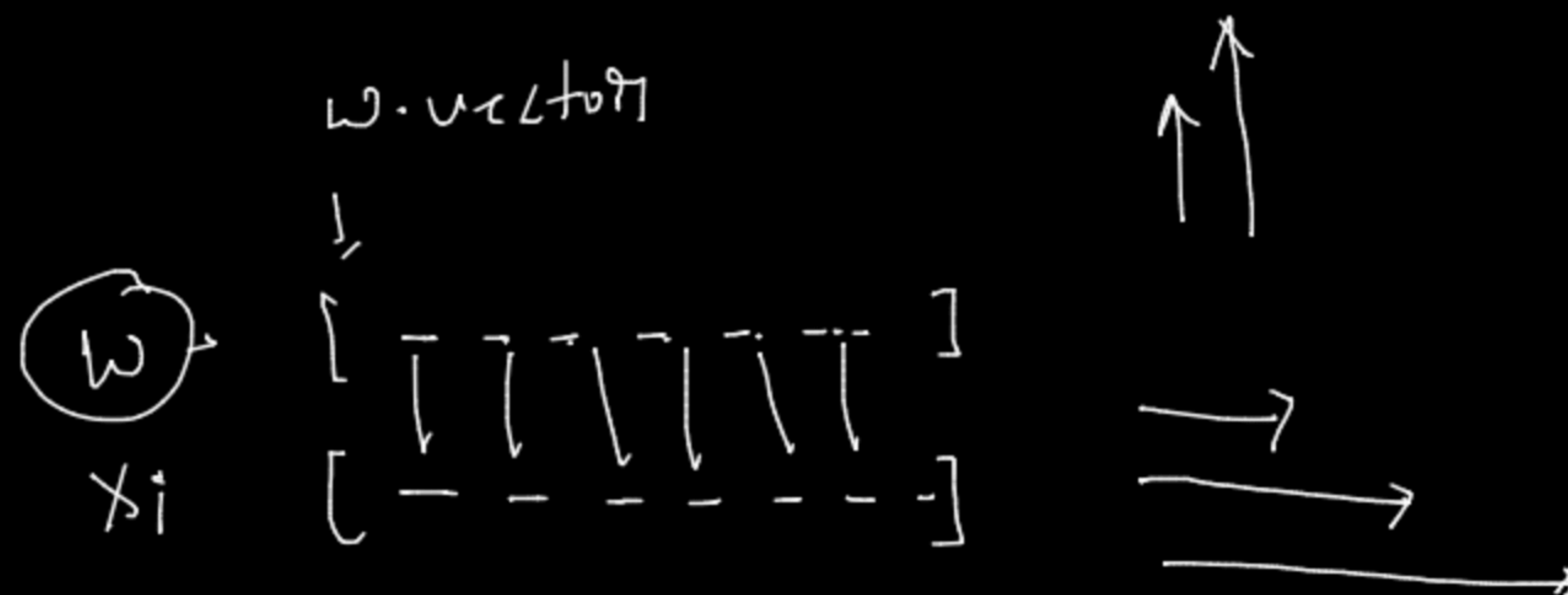
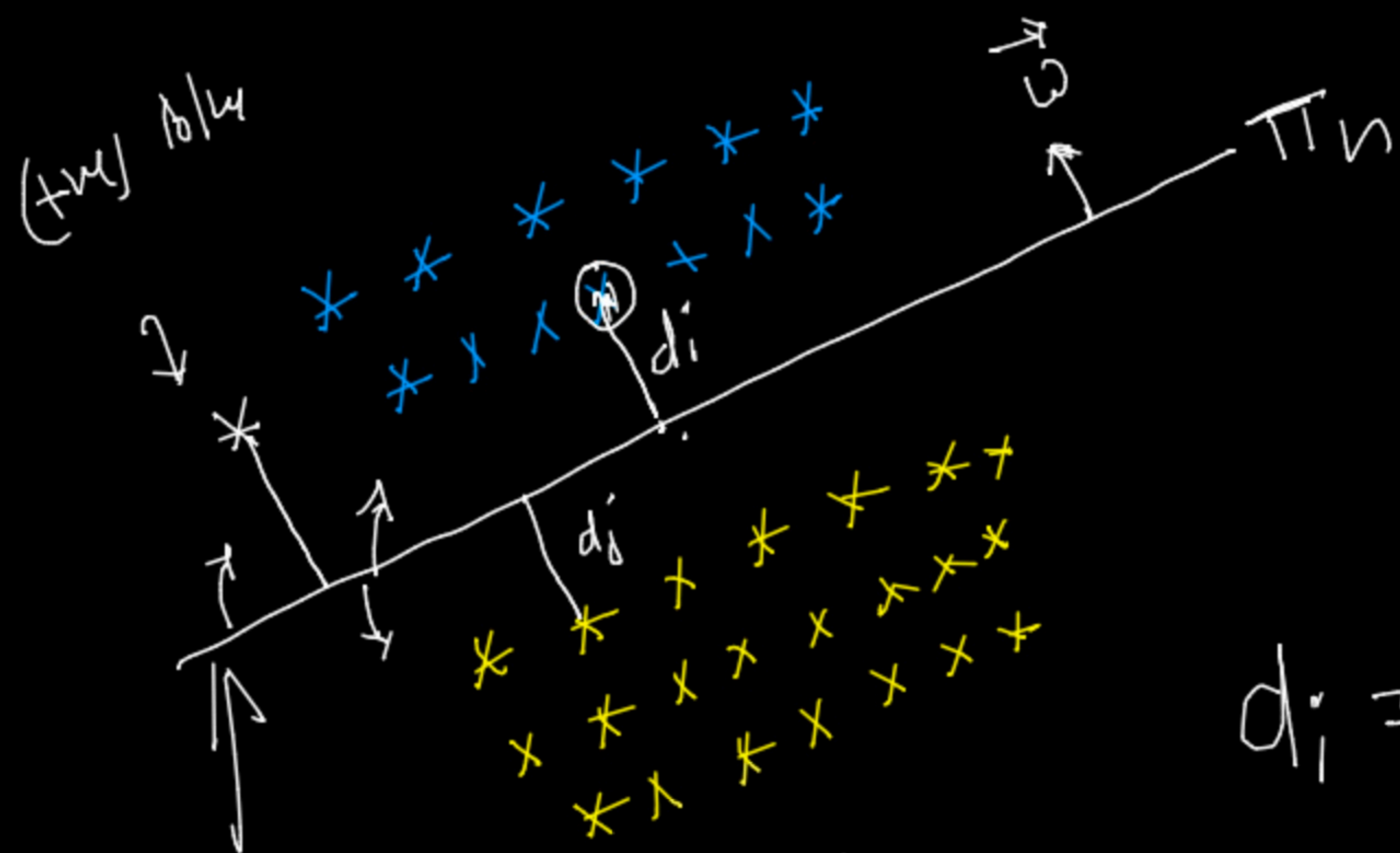
$$Q_n \Rightarrow \{x; y\}$$

Task find out

w & b

such that it separates pos pts & neg pts with
maximum distance





$$d_i = \frac{w^T \cdot x_i}{\|w\|}$$

identity vector $\rightarrow \textcircled{1}$

$$\hat{w} = \frac{w}{\|w\|}$$

'Decision' surface

if $\boxed{\|w\| = 1}$

$$\begin{cases} d_i = w^T \cdot x_i \rightarrow \textcircled{+w} \\ d_j = w^T \cdot x_j \rightarrow \textcircled{-w} \end{cases}$$



Classifier say decision surface

if $y \cdot w^T \cdot x_i > 0$
 $y_i \rightarrow (+ve)$ Yes

if $w^T \cdot x_i \neq 0$
 $\rightarrow y_i = (-ve)$

$y_i \cdot w^T \cdot x_i > 0$
 Actual

$y_i \cdot w^T \cdot x_i > 0$
 $\rightarrow$ correct
 $\rightarrow$ classification

$(+ve) \cdot (+ve) > 0$
 $\rightarrow$ correct classification

$y_i \cdot w^T \cdot x_i$
 $(-ve) \cdot (-ve)$
 $\downarrow$
 $(+ve)$

$(-ve) \cdot (+ve)$
 $\rightarrow$ wrong
 $(+ve) \cdot (-ve)$
 $\rightarrow$ wrong

sklearn

$y_i \cdot w^T \cdot x_i$
 $(-ve) \cdot (-ve)$

$(+ve)$

x_1	$+ve$
x_2	$+ve$
x_3	$-ve$
x_4	$-ve$

$$d = w^T \cdot x_i$$

max

argmax d

w

↓ mod

signed distance

$(+w, -w)$

optimization problem

$$\text{arg-max} \left(\sum_{i=1}^n y_i \cdot \underline{w^T \cdot x_i} \right)$$

w^T

Data

①